

Polynomials – 9

Cyclotomic polynomials and the arithmetic of roots of unity

Homework

24 сентября 2026 года

Theoretical minimum. Cyclotomic polynomials Φ_n : integer coefficients, $x^n - 1 = \prod_{d|n} \Phi_d$, the Möbius formula $\Phi_n = \prod_{d|n} (x^d - 1)^{\mu(n/d)}$, the values $\Phi_n(1)$, the formulas for Φ_p and Φ_{pq} ; irreducibility over $\mathbb{Q}$, so Φ_n is the minimal polynomial of a primitive n -th root of unity ζ and $\sum a_k \zeta^k = 0$ ($a_k \in \mathbb{Q}$) iff $\Phi_n \mid \sum a_k x^k$; non-negative vanishing sums of pq -th roots are unions of regular p - and q -gons, and the Lam–Leung theorem. Over $\mathbb{F}_p$: the power sums $\sum x^k$, polynomial functions of degree $< p$, Hermite’s criterion for permutation polynomials. Annihilating ideals of the shift operator $f(x) \mapsto f(x+h)$ are principal and closed under gcd.

1. Prove the formula for $\sum_{x \in \mathbb{F}_p} x^k$ and Hermite’s criterion. Deduce that x^k permutes $\mathbb{F}_p$ if and only if $\gcd(k, p-1) = 1$, and that no polynomial of degree $d > 1$ with $d \mid p-1$ is a permutation polynomial of $\mathbb{F}_p$.
2. Let p be a prime. Prove that $\Phi_{pn}(x) = \Phi_n(x^p)$ if $p \mid n$, that $\Phi_{pn}(x) = \Phi_n(x^p)/\Phi_n(x)$ if $p \nmid n$, and that $\Phi_{2n}(x) = \Phi_n(-x)$ for odd $n > 1$. For distinct primes p, q prove that all coefficients of Φ_{pq} lie in $\{-1, 0, 1\}$ by expanding $\frac{1-x}{(1-x^p)(1-x^q)}$ modulo x^{pq} , and show that the coefficient of x^7 in Φ_{105} equals -2 .
3. (Kronecker) Let $f \in \mathbb{Z}[x]$ be monic with $f(0) \neq 0$, all of whose complex roots satisfy $|z| \leq 1$. Prove that all roots of f are roots of unity. (Consider the polynomials $f_m = \prod (x - z_i^m)$.) Deduce that an algebraic integer all of whose conjugates have absolute value 1 is a root of unity, and show by the example $\frac{3+4i}{5}$ that integrality cannot be dropped.
4. Find all pairs of cubical dice whose faces are labelled by positive integers (repetitions allowed) such that the sum of the two numbers thrown has

the same distribution as for two ordinary dice. (Factor the generating function $(x + x^2 + \cdots + x^6)^2$ into cyclotomic polynomials.)

5. A family of at least two sets is called a partition of a set X if the sets are pairwise disjoint and their union is X .
 - a) Construct a partition of the set of all positive integers into infinitely many disjoint infinite arithmetic progressions with different common differences.
 - b) Can the set of all positive integers be partitioned into a finite number of such progressions?

(ISTCiM 2022)

6. Determine all pairs $P(x), Q(x)$ of complex polynomials with leading coefficient 1 such that $P(x)$ divides $Q(x)^2 + 1$ and $Q(x)$ divides $P(x)^2 + 1$.
(IMC 2018)

7. Let m be a positive integer and let p be a prime divisor of m . Suppose that the complex polynomial $a_0 + a_1x + \cdots + a_nx^n$ with $n < \frac{p}{p-1}\varphi(m)$ and $a_n \neq 0$ is divisible by the cyclotomic polynomial $\Phi_m(x)$. Prove that there are at least p nonzero coefficients a_i .
(VJIMC 2015)

8. Suppose that for a function $f: \mathbb{R} \rightarrow \mathbb{R}$ and real numbers $a < b$ one has $f(x) = 0$ for all $x \in (a, b)$. Prove that $f(x) = 0$ for all $x \in \mathbb{R}$ if

$$\sum_{k=0}^{p-1} f\left(y + \frac{k}{p}\right) = 0$$

for every prime number p and every real number y .
(IMC 2010)

9. Let p be an odd prime. Show that for at least $(p+1)/2$ values of n in $\{0, 1, 2, \dots, p-1\}$ the number

$$\sum_{k=0}^{p-1} k! n^k$$

is not divisible by p (here $0^0 = 1$).
(Putnam 2011 B6)

10. Let k and n be positive integers with $n \geq k^2 - 3k + 4$, and let

$$f(z) = z^{n-1} + c_{n-2}z^{n-2} + \cdots + c_0$$

be a polynomial with complex coefficients such that $c_0c_{n-2} = c_1c_{n-3} = \cdots = c_{n-2}c_0 = 0$. Prove that $f(z)$ and $z^n - 1$ have at most $n - k$ common roots.
(IMC 2017)